Atomic elementary flux modes explain the steady state flow of metabolites in large-scale flux networks

aEFMs explain metabolite flows

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Flux network/Elementary flux mode/flow decomposition/Markov chain/atom mapping

Abstract

Steady state fluxes are a measure of cellular activity under metabolic homoeostasis, but understanding how individual substrates are metabolized remains a challenge in large-scale networks. Pathway-based approache[illegible]lux mode (EFM) analysis are limited to small networks due to the comb[illegible]thways and ambiguity decomposing fluxes onto EFMs. Here, we present a[illegible]ch to explain metabolic fluxes in terms of the steady state flow of their molecular constituents. We refer to these pathways as atomic EFMs (AEFMs) and show that computations involving AEFMs are orders of magnitude faster than standard EFMs which balance molecular stoichiometries. Using our approach, we enum[illegible]gen AEFMs in five genome-scale metabolic models and compute the AEFM[illegible]es-timated in a HepG2 liver cancer cell line. Our results systematically cha[illegible]bolite remodelling and reveal the most active pathways of HepG2 glutamine metabolism. We anticipate our method will be useful for quantifying cellular metabolism as high-throughput fluxomic analyses become increasingly common.

atomic

Say something more biological! "weird" TCA...

Introduction

Recent technological and computational advancements are leading to a rise in metabolic flux data (Lee et al, 2019; Wang et al, 2022a,b). These fluxes quantify the rate of metabolite interconversions within a cell or their movements across subcellular compartments. Metabolic fluxes are studied to understand the dynamics of substrate switching (Zampieri et al, 2019; Kochanowski et al, 2021), redox balancing (McKinlay & Harwood, 2010; Franchina et al, 2022), overflow metabolism (Basan et al, 2015; Vander Heiden et al, 2009), cell differentiation (Endo et al, 2022; Jackson & Finley, 2024), and cellular communities (Schuetz et al, 2012; Yu et al, 2022; Gustafsson et al, 2024). Hence, 'fluxomics' (Emwas et al, 2022; Sanford et al, 2002) is emerging as a field dedicated to quantifying the totality of fluxes within biological systems.

Current experimental and computational methods generally estimate fluxes under metabolic homeostasis. Under this assumption, the production and consumption of metabolites are balanced over time, and the fluxes are said to be under steady state. Techniques such stable isotope tracing analysis (SITA) experimentally determine steady state fluxes by culturing cells in labelled substrates and quantifying their incorporation within endogenous metabolites (Buescher et al, 2015; Antoniewicz, 2018). Using this method, fluxes have been measured in major metabolic subsystems such as glycolysis, pentose phosphate pathway, and citric acid cycle (Orth et al, 2010b; Ahn et al, 2017; Wu et al, 2019). Many computational methods have also been proposed to infer steady state fluxes from transcript, protein, and metabolite abundance data (Uematsu et al, 2022; González-Arrué et al, 2023; Huang et al, 2023; Kaste & Shachar-Hill, 2023; Lee et al, 2023). Collectively, these approaches have been used to estimate and predict fluxes in genome-scale metabolic models (GEMs) involving hundreds to thousands of reactions (Nilsson et al, 2020b; Cherkaoui et al, 2022; Gelbach et al, 2024).

Although much effort has been made to estimate steady state fluxes, less emphasis has been placed on developing methods to analyze these large-scale flux datasets. In contrast to abundance data (e.g. transcripts, proteins, metabolites), steady state fluxes are constrained by the reaction stoichiometries between metabolite inflows and outflows. In a network with fixed source and sink fluxes an increase in flux along one metabolic pathway must decrease fluxes along one or more other pathways to maintain mass conservation. For simple networks, one may predict how the system would evolve by identifying the up and downstream reactions flanking an enzymatic perturbation. For example, Schwartz & Kanehisa were able to enumerate all glycolytic pathway fluxes in a 12 reaction model of yeast glycolysis (Schwartz & Kanehisa, 2006). Yet in more complex genome-scale metabolic networks containing thousands of fluxes (e.g. Brunk et al (2018); Robinson et al (2020); Wang et al (2021)), there exists no hierarchical relationship between reactions. A change in one reaction rate, let alone multiple perturbations, may alter network fluxes beyond neighbouring reactions (Nobile et al, 2021).

This problem of understanding the flow of metabolites within steady state flux networks has historically been addressed through the concept of elementary flux modes (EFMs). EFMs are a pathway-based approach to analyze metabolic networks and are defined as minimal sets of biochemical reactions that maintain steady state flux (Schuster & Hilgetag, 1994). This minimal property specifies that an EFM cannot be decomposed into two or more smaller pathways carrying steady state flux. An attractive property of EFMs is that any set of steady state fluxes in a network can be explained as a positive, linear combination of EFMs (Schuster et al, 1999). While EFMs are a mechanistic explanation of how metabolites travel through networks, the number of EFMs scales combinatorially as a function of network size (Acuña et al, 2009). It remains computationally infeasible to enumerate EFMs in GEMs after over a decade of algorithmic advancements (e.g. Gagneur & Klamt 2004; Terzer & Stelling 2008; Buchner & Zanghellini 2021). The number of EFMs also poses

a challenge for explaining network fluxes in terms of their EFMs. When there are more EFMs than linearly independent fluxes, there exist multiple ways to assign EFM weights that reconstruct the observed network fluxes. Consequently, structural analyses of EFMs and downstream analyses of their weights have been limited to small-scale metabolic subnetworks (Fakih et al, 2023; Mpabanzi et al, 2022; Toya & Shimizu, 2022; Matsuda et al, 2011; Schwartz & Kanehisa, 2006).

In our previous work, we took the first steps to solve the EFM flux decomposition problem for the special case of unimolecular reaction networks under a Markov constraint. We modelled a hypothetical particle moving between metabolites under the assumption that the transition probability of each reaction was proportional to the observed steady state flux. By expanding the notion of Markov chain state to include sequences of metabolites, we proposed an algorithm to efficiently compute simple cycle probabilities corresponding to a given EFM. We called this construct the *cycle-history Markov chain* (CHMC) and proved that these EFM probabilities could be scaled to weights that fully reconstructed the flux along each reaction (Chitpin & Perkins, 2023).

In this work, we generalize our CHMC method to operate on any type of metabolic model including those involving multispecies reactions. We do so by proposing a new type of *atomic elementary flux mode* which traces the steady state flow of an individual atom within a metabolic network. We refer to these as AEFMs to distinguish them from the (molecular) EFMs proposed by Schuster & Hilgetag. While a linear combination of molecular EFMs will reconstruct the overall reaction fluxes, the totality of AEFM weights for a given source metabolite will always reconstruct its mass flow entering the network. We first introduce our framework for enumerating and uniquely identifying AEFM weights. Using a state-of-the-art atom mapping algorithm, we show how to generate atomic transition graphs from which we construct atomic cycle-history Markov chains (ACHMCs). From these ACHMCs, we enumerate AEFMs in GEMs considered infeasible by state-of-the-art molecular EFM enumeration programs. We further analyse the structure of the AEFMs to

characterize nutrient source remodelling. Lastly, we analyze a single set of AEFM weights estimated from a HepG2 liver cancer cell line to quantify glutamine metabolism. Our results predict glutamine in HepG2 cells is metabolized through a non-canonical TCA pathway recently discovered in non-small cell lung cancer (NSCLC) cells and C2C12 mouse myoblasts (Arnold et al, 2022).

Results

Tracing atomic flows in genome-scale models

We first describe what molecular EFMs look like in metabolic networks and provide intuition why enumerating and decomposing fluxes onto these pathways remains a fundamental problem in literature. Fig 1a shows an example molecular EFM which shows how the substrates glucose and ATP are consumed to generate product metabolites G3P, DHAP, and ADP. Multiple source and sink fluxes are required to balance the internal metabolite stoichiometries, leading to several branching and converging paths within the molecular EFM. In larger metabolic networks, molecular EFMs can become increasingly complex with numerous source and sink fluxes required to stoichiometrically balance participating metabolites with their associated energetic molecules (e.g. ATP/ADP), redox equivalents (e.g. NAD^+/NADH), metabolic byproducts (e.g. water, hydrogen ions) and coenzymes (e.g. coenzyme A). This complexity can lead to an intractable number of pathways that may only differ by reaction subsets required to mass-balance these metabolites. Choosing metabolic byproducts to remove may, in part, solve this problem. However, the choice of metabolites to prune from the network is often subjective and not guaranteed to sufficiently reduce the total number of molecular EFMs.

Given many metabolic flux networks are generated to study nutrient catabolism (Zamboni et al, 2015), we argue for studying source metabolite-derived AEFMs to reduce the combinatorial number

of molecular EFMs. An example carbon AEFM is highlighted in Fig 1b which shows how the C6 glucose carbon is remodelled during the first steps of glycolysis. As atoms are never split apart or fused together in metabolic reactions, these AEFMs always correspond to simple paths spanning source and sink metabolites or simple cycles between internal metabolites. These pathways are straightforward to interpret because each atom within an AEFM can only transit from a single substrate to product, regardless of the network size, connectivity, or number of participating substrates/products in each reaction. This property further enables us to enumerate and compute the AEFM weights using our previously described CHMC method which can uniquely decompose fluxes onto EFMs for unimolecular reaction networks (Chitpin & Perkins, 2023).

kill "optional"

Fig 1c outlines our pipeline to enumerate and identify AEFM weights. In inputs the metabolic network defined by a set of metabolites and reactions enco try matrix, the metabolite structures denoted as SMILES strings, and (optionally) the steady state fluxes. From the stoichiometry matrix and SMILES strings, we construct reaction SMILES strings which are imputed into the atom mapping program RXNMapper (Schwaller et al, 2021) to trace the position of each atom within each reaction equation. This atom mapping data is used to construct an atomic transition graph in step 2, which traces the atomic position of an individual atom from a given source metabolite entering the network. We set the transition probabilities equal to the atomic flux through that reaction. For example, 10mM of input glucose flux would result in 10mM of atomic flux through each glucose carbon. From each atomic transition graph, we construct the corresponding ACHMC to enumerate all AEFMs involving that source metabolite atom in step 3. Finally, the AEFM weights are computed in step 4 by steady state analysis of the ACHMC. If no steady state fluxes are imputed, only the enumerated AEFMs in step 3 are returned.

Computing AEFMs is computationally tractable in large-scale networks

Given the major computational challenge o[illegible]r EFMs in networks with > 100 metabolites and reactions, we first sough[illegible]s feasible to enumerate atomic EFMs in GEMs. We selected five genom[illegible]oss distinct organisms in ascending order of network size. These networks included a metabolic model of *E. coli*, human red blood cells, *H. pylori*, *S. aureus*, and HepG2 liver cancer cells. Pseudometabolites and pseudoreactions were removed from these metabolic models to ensure unambiguous atom mappings for each reaction. After model pre-processing, the resulting networks ranged between 71-547 metabolites and 144-974 reactions. We then attempted to enumerate the molecular EFMs using FluxModeCalculator which is the fastest molecular EFM enumeration program (van Klinken & Willems van Dijk, 2016). Following our pipeline, we subsequently compiled the metabolite SMILES strings and constructed ACHMCs rooted on each carbon and nitrogen atom across all source metabolites. We focused on carbon and nitrogen atoms, in particular, for their metabolic significance over chemical elements such as oxygen or phosphorus. The resulting summary statistics describing the GEMs and ACHMCs are presented in Table 1. Across all carbon ACHMCs, we observed a broad increase in the total number of AEFMs as a function of metabolic network size. The one exception to this finding was the HepG2 network which contained fewer metabolites and reactions than iSB619, yet exhibited two orders of magnitude more carbon AEFMs which suggested a more highly-interconnected network (Fig EV1). Collapsing across carbon AEFMs and nitrogen AEFMs, we observed a consistent ratio between the number of ACHMC transitions and ACHMC states ($\mu_C = 1.43, \sigma_C^2 = 0.14$; $\mu_N = 1.53, \sigma_N^2 = 0.16$).

Fig 2a shows the total number of atomic and molecular EFMs enumerated by our program MarkovWeightedEFMs.jl and FluxModeCalculator on the five GEMs. Using our ACHMC pipeline, we enumerated between $10^3 - 10^6$ carbon and nitrogen EFMs within each GEM. In contrast, Flux-

Consider: Because it computes a different notion of pathway (AEFM versus molecular EFM), our approach
OR alter intro sentence as "Because the number of AEFMs is so much smaller than number of EFMs..."

M[illegible] nallest E. coli core and iAB[illegible] atomic EFMs.

By focusing on an a[illegible] approach g[illegible] d the number of EFMs to biologically relevant pathways involving nutrient source remodelling. Our approach was furthermore significantly faster than the state-of-the-art program FluxModeCalculator. Our benchmarking showed ACHMC computation times to be $10^3 - 10^4$ tim[illegible] lator running in serial. Running FluxModeCalculator in pa[illegible] ove run times by 8.2 8.6 times (Fig EV2), [illegible] corresponding network carbon and nit[illegible]

3 paragraphs too much for computation time considerations?

These fast AEFM enumeration tim[illegible] atom positions is much larger than the[illegible] To resolve these findings, we enumerated [illegible] nitrogen positions across all five GEM[illegible] ted the greatest number of combinations wi[illegible] metabolite-carbon and 2555 metabolite-nitrogen positions. This corresponded to over 18 and 4 times the number of metabolites within each respective network. However, each carbon or nitrogen atomic transition graph is not necessarily constructed from this entire list of metabolite-atom positions. Constrained by the atom mapping predictions, we found that the average carbon and nitrogen ACHMC acr[illegible] ly transited 0.41% and 1.7% of the total number of metabolite-carbon and me[illegible] espectively (Fig EV3). Hence, the metabolite-atom state space of any given [illegible] ders of magnitude smaller than the total number of network metabolites.

Branching factor? Think!

We hypothesized that the speedup between MarkovWeightedEFMs.jl and FluxModeCalculator was therefore due to the proportionally fewer number of AEFMs-versus-molecular EFMs across the metabolic networks. For molecular EFM enumeration algorithms, it is well established that run

times cannot be predicted from the number of metabolites or reactions within the model (Ullah et al. [illegible] AEFM run times scale approximately quadratically as a function of th[illegible]Cs (Fig 2c; slope $= 1.90$, intercept $= -6.54$). While future work is requi[illegible] algorithm scaling, we emphasize that the computation of each ACHMC is embarrassingly parallel at the source metabolite-atom level. Our software is also parallelizable on the level of individual ACHMCs to further speed up AEFM enumeration. For those top 28 largest carbon ACHMCs in the HepG2 network, we re-enumerated their AEFMs with varying numbers of threads. Fig 2d reports the algorithm scaling, where we observed a median speedup of 7.1 times when using 32 threads. As expected, the greatest speedups were associated with the largest ACHMCs. For example, the largest ACHMC corresponded to a glutamine carbon with 1,120,290 ACHMC states/168,793 AEFMs, which exhibited a 9.3 times speedup when enumerated with 32 threads.

can be parallelized at the...

Most AEFMs are source-to-sink pathways spanning dozens of reactions

Having systematically enumerated all carbon and nitrogen EFMs across the five GEMs, we next turned to characterizing their structural properties. Molecular EFMs can be divided into pathway that span source and sink metabolites or internal loops within the metabolic network such as pairs of reversible reactions. AEFMs can be further classified into those that transit sets of distinct metabolites or those that revisit the same metabolite, albeit in different atomic positions. An example of a common source-to-sink atomic EFM with metabolite revisitations is a carbon AEFM tracing glucose remodelling through two turns of the TCA cycle; on the second turn, the position of the carbon atom shifts within the TCA intermediates, resulting in different isotopomers and therefore a metabolite revisitation under our definition.

To obtain a broad understanding of the types of AEFMs within the five GEMs, we categorized

them into source-to-sink pathways and internal cycles with or without metabolite revisitations in different atomic positions. Fig 3a-b shows the counts for each AEFM classes within each GEM. We observed that the majority of carbon AEFMs were source-to-sink pathways with or without metabolite revisitations. The number of these pathways without metabolite revisitations tended to scale as a function of metabolic network size. The number of looped carbon AEFMs without metabolite revisitations increased slightly while those with revisitations varied across the five GEMs. Similar findings were observed for the nitrogen AEFMs, although looped pathways with metabolite revisitations were only observed in the iIT341 and iSB619 networks. The iAB RBC 283 dataset was unique in that it only contained source-to-sink and looped pathways with no metabolite revisitations. This is likely due to the lack of cellular organelles in red blood cells and therefore no cyclic reactions associated with metabolite transport between compartments or mitochondrial TCA activity.

Fig 3c-d shows the distribution of atomic EFM lengths across the five GEMs. The carbon AEFMs from all but the HepG2 networks tended to be quite short with few pathways exceeding 50 reactions. In contrast, the HepG2 carbon AEFM lengths exhibited a Gaussian distribution with mean of 61.7 and variance of 15.5. As there is a greater number of metabolites and reactions in the iSB619 network, these results suggest a higher degree of connectivity between metabolites within the HepG2 network. In particular, to emphasize how different the paths available to each atom in a metabolite may be, we computed the number of AEFMs for each carbon atom (resp. nitrogen atom). For each metabolite, we then computed the ratio of the maximum number of paths for any carbon (resp. nitrogen), and the minimum number of paths. So for example, a ratio of 10 means that one of the carbon atoms can transit the network in 10 times as many different ways as the carbon with the fewest paths through the network. Fig 3e-f shows that a surprising number of carbons and nitrogens can transit different pathways within their respective source metabolites. Specifically, many hydrophobic and positively charged amino acids such as phenylalanine, methionine, arginine,

and histidine could undergo carbon and nitrogen ratios over 10^2.

A minority of carbon AEFMs explains the majority of source metabolite remodelling

We finally highlight the utility of our AEFM method by analyzing a single set of steady state fluxes in the HepG2 dataset. Specifically, we focused on identifying alternative pathways of glutamine metabolism as many cancer cells, including HepG2 cells, uptake more glutamine than is required for biomass production (Ye et al, 2023; Yoo et al, 2020). Using our method, we decomposed the set of steady state fluxes onto the 241,085 glutamine carbon AEFMs characterizing the HepG2 metabolic network and sought to identify the most dominant pathways carrying the greatest glutamine carbon flow through the network. Fig 4a shows the cumulative explained mass flow of each glutamine carbon as a function of their AEFMs ranked from greatest to smallest explained mass flow. We capped the cumulative explained mass flow of each glutamine carbon at 99% because we observed that hundreds to thousands of carbon AEFMs collectively explained the remaining 1% glutamine mass flow. The resulting curves indicate that the majority of glutamine carbon remodelling is explained by a surprisingly small number of glutamine AEFMs. The top 5 ranked AEFMs across each glutamine carbon explained over half (50.2%) of the total glutamine carbon mass flow with the subsequent 5 next highest ranked AEFMs only explaining an additional 9.8% glutamine carbon mass flow. These observations were consistent across carbon AEFMs derived from other amino acids as well as glucose (Fig EV3). For some source metabolites, we observed overlapping curves between carbons (e.g. arginine, glycine, serine), suggesting individual carbons were remodelled through the same AEFMs. Corroborating results from Fig 3-e-f, we observed many non-overlapping curves for the majority of amino acids where their carbons could transit distinct pathways involving different

internal metabolites.

To obtain a high-level understanding of the dominant pathways of glutamine carbon remodelling, we constructed a metabolic subnetwork from the top 5 glutamine carbon AEFMs, collapsing metabolite-atom positions onto distinct metabolites. Of the 435 metabolites in the HepG2 dataset, input glutamine can remodel into 202 species. We found that the top 5 glutamine-derived carbon AEFMs consist of just 23 metabolites and 31 reactions (Fig 4b). These AEFMs mostly consisted of cyclic pathways involving TCA intermediates. Only two source-to-sink reactions converting cytosolic glutamine towards the protein pool and carbon dioxide through alpha-ketoglutarate in the mitochondria.

As AEFMs are steady state pathways, they can be visualized as Sankey diagrams to better understand the pathway fluxes. Fig 4c-e shows the top 5 AEFMs for each glutamine carbon with metabolites represented as nodes and each AEFM as a flow proportional to its weight. Although each glutamine carbon can transit a different number of AEFMs, we found the top 5 backbone carbons and first two sidechain carbons were identical in both AEFM pathway and weight.

Focusing on either backbone carbons in Fig 4c, we observed that nearly all AEFMs corresponded to well-established metabolic subsystems. Both the first and third highest ranked AEFMs corresponded to all metabolites and reactions involved in the malate-aspartate shuttle. Nearly all of the metabolites in the fourth largest AEFM matched canonical TCA cycle with two additional reactions between alpha-ketoglutarate transforming into succinyl-CoA. We noticed that the second largest AEFM cycled carbon mass between TCA intermediates citrate, oxaloacetate, and malate between the cytosol and mitochondria. Interestingly, this metabolic pathway corresponds to the recently discovered non-canonical TCA activity described by Arnold et al, who validated this pathway by stable isotope tracing analysis in a different (non-small cell lung) cancer cell line. This non-canonical TCA activity was also observed in AEFM 2 in Fig 4e among the glutamine carbons 3 and 4. We

also found AEFMs 1, 3, and 5 in Fig 4d and AEFMs 3 and 5 in Fig 4e also corresponded to the malate-asparate shuttle. Only the terminal glutamine carbon in Fig 4e contained source-to-sink AEFMs that decarboxylated glutamine-derived alpha-ketoglutarate into carbon dioxide.

Having mapped each AEFM to an existing metabolic subsystem, we asked whether we could combine the results across all Sankey diagrams into a simplified metabolic model describing bulk glutamine remodelling in the HepG2 network. The resulting schematic is shown in Fig 4f which consists of the following three major metabolic pathways: canonical TCA cycle, non-canonical TCA cycle, and the malate-aspartate shuttle. These metabolic pathways were chosen based on the number of top glutamine-derived carbon AEFMs that overlapped with reactions within these subsystems.

Our results suggest input glutamine is converted into cytosolic glutamate which is transported into the mitochondria through aspartate/glutamate antiporters encoded by *SLC25A12* and *SLC25A13*. Mitochondrial glutamate is then converted into TCA intermediate alpha-ketoglutarate and citrate. Instead of following the canonical reactions of the TCA cycle, the mitochondrial citrate is exported to the cytosol where it can participate in non-canonical TCA cycle activity or the malate-aspartate shuttle to re-enter the mitochondria. Taken together, the dominant carbon AEFMs predicted by our model suggest HepG2 glutamine is conserved within TCA intermediates that are regenerated by shuttling mitochondrial citrate to the cytosol.

Discussion

Since the proposal of molecular EFMs, much effort has been made to enumerate and decompose steady state metabolic fluxes onto these pathways with little success. By defining EFMs with respect to individual atoms rather than molecules, we introduced the concept of AEFMs which describe how

individual atoms are remodelled within a metabolic network. As AEFMs never involve branching pathways, they are easier to interpret than their molecular EFM counterparts which may involve multiple inputs and outputs to balance molecular stoichiometries. Using the atom mapping program RXNMapper, we adapted our previously described CHMC method to efficiently enumerate carbon and nitrogen AEFMs in five GEMs. Our structural analysis systematically characterized source metabolite remodelling at the atomic level. By comparing the number of AEFMs across atoms within the same source metabolite, we revealed how certain amino acids displayed remarkable variability in the number of pathways transited by their constituent atoms. AEFM weight analysis of the HepG2 network revealed that the majority of glutamine carbons were remodelled via cyclic pathways regenerating TCA intermediates through cytosolic citrate. Excitingly, our model also predicted non-canonical TCA activity which has not been previously described in HepG2 cells.

As a systematic method to explain the flow of metabolites, the problem of molecular EFM enumeration has historically limited their applicability to small-scale networks. Many algorithmic advances enumeration have incrementally improved molecular EFM enumeration, yet never feasibly beyond networks with > 100 metabolites and reactions (Kamp & Schuster, 2006; Terzer & Stelling, 2008; van Klinken & Willems van Dijk, 2016). Molecular EFM enumeration is generally regarded as unfeasible for large-scale networks—so much so that numerous methods have been developed to enumerate a subset of molecular EFMs constrained by pathway length (de Figueiredo et al, 2009), reaction thermodynamics (Gerstl et al, 2015), participating metabolites (Pey & Planes, 2014), participating reactions (Kaleta et al, 2009), gene regulatory rules (Jungreuthmayer et al, 2013), metabolic subsystems (Schuster et al, 2002; Schwartz et al, 2007), extracellular flux measurements (Soons et al, 2011; Jungers et al, 2011), or even random sampling (Machado et al, 2012). While our HepG2 results did show the majority of source metabolite fluxes were explained by a small number of AEFMs, contextualizing the significance of these dominant pathway fluxes is

only possible when all AEFMs are known. Furthermore, reaction thermodynamics, gene regulatory rules, and other constraints may change across fluxes estimated from different biological conditions. Our results suggest that enumerating the shortest pathways would fail to recover the majority of TCA pathways, as the top five ranked AEFMs across all glutamine carbons averaged 6 reactions in length. By computing AEFMs, instead, we could feasibly enumerate atomic pathways in genome-scale networks containing hundreds of metabolites and reactions within minutes or hours on desktop computing resources. We are hopeful that algorithmic tricks borrowed from molecular EFM enumeration tools, such as linear pathway compression, could further decrease memory usage and run times.

Aside from carbon and nitrogen AEFMs, our work is broadly applicable to other types of chemical elements. One could feasibly enumerate oxygen or hydrogen EFMs in small-scale networks, although the significance of these pathways is unclear for biochemical networks given the ubiquitous nature of these atoms. We also caution against enumerating hydrogen AEFMs, in particular, given limitations associated with RXNMapper. Unless explicitly represented in the reaction SMILES string, RXNMapper does not assign hydrogen atom mappings. While this may seem trivial, RXNMapper cannot map atoms in reaction SMILES strings longer than 512 characters. Our method could also be applied to study sulfur metabolism to quantify methionine or cysteine remodelling, or phosphorus metabolism to quantify nucleotide synthesis. Outside the realm of biochemistry, our method is applicable to any periodic table element and possibly relevant for studying general chemical reaction networks. While statistical analyses, such as different AEFM weight analysis, may reveal metabolic changes in individual pathways altered across conditions, it is likely that changes in a metabolic pathway may impact weights in multiple AEFMs. For example, the majority of AEFMs across distinct source metabolites are not "molecularly" independent. By this, we imply that some AEFMs may correspond to the same sequence of metabolites or different metabolite/atom sequences

stemming from shared reactions. Their weights will likewise be related through these common reactions. This problem may perhaps be addressed by formalizing a relationship between atomic and molecular EFMs. While we have taken significant steps towards solving the AEFM enumeration and flux decomposition problem, we expect rapid computational interest analyzing AEFM weights in the near future.

Materials and Methods

Reagents and Tools table

See tables/reagents-and-tools-table.xls.

Methods and Protocols

Datasets

Five highly-curated GEMs were chosen in this study to reflect a diversity of organisms, metabolites and biochemical reactions. Four networks were obtained from the BiGG UCSD database (King et al, 2016) in ascending order of network size. Datasets were downloaded as SBML (.xml) files, and the corresponding stoichiometry matrices, metabolites and reactions were extracted using the Julia SBML.jl package. The HepG2 liver cancer cell line dataset was chosen for its single set of estimated steady state fluxes estimated from a combination of protein and metabolite abundance data. These cells were grown in DMEM with 10% fetal calf serum, 22mM glucose, and 1.8mM glutamine (Nilsson et al, 2020b). The steady state fluxes were recomputed from the flux balance analysis (FBA) model provided by the authors (figure3_metabolitebalances.m; MATLAB R2018a) (Nilsson et al 2020b; Data ref: Nilsson et al 2020a). Across all models, we modelled all reversible reactions as net

forward reactions to simplify the number of molecular and AEFMs as the majority of experimental and computational methods can only estimate net fluxes. We assumed the reaction stoichiometries of the BiGG datasets represented the canonical net directions of all reversible reaction. For the HepG2 dataset, the net directions were chosen based on the net fluxes estimated from the FBA model.

Extracting SMILES strings

SMILES structures, unfortunately, were not encoded in the BiGG SBML files nor the HepG2 dataset. We manually extracted SMILES strings corresponding to each metabolite across all GEMs. Isomeric SMILES strings were taken from PubChem (Kim et al, 2023), the Human Metabolome Database (Wishart et al, 2022), or MetaNetX database (Moretti et al, 2021). Metabolites shared across GEMs were assigned the same SMILES structure in addition to identical metabolites stratified across distinct subcellular compartments. Generic structures such as R-groups were not modelled in this study and were replaced with a single bonded hydrogen under the assumption this modification would not alter atom mappings, especially since RXNMapper does not map hydrogen atoms unless they are explicitly represented in the SMILES strings. SMILES strings were also not assigned to metabolites with ambiguous structures nor pseudometabolites with no known chemical structure. All pseudometabolites and reactions involving pseudometabolites were removed from the stoichiometry matrix (Table 2).

Constructing reaction SMILES strings

RXNMapper performs atom mapping on a string that encodes the SMILES structures of all substrates and products for a given reaction. These reaction SMILES strings are constructed by concatenating SMILES strings of individual substrates and products. Multiple substrates or prod-

ucts are separated by the symbol '.', while the reaction arrow delimiting substrates and products is denoted by the symbol '¿¿'. The substrate and product SMILES strings are concatenated in order of metabolite appearance within the stoichiometry matrix, and multiple stoichiometric copies are consecutively repeated in the reaction SMILES strings. Non-integer stoichiometric coefficients are therefore not allowed and these pseudoreactions were removed from the stoichiometry matrix (Table 2). We manually checked several reactions to ensure the assigned atom mappings were correct. Crucially, we found RXNMapper to fail on nearly all classes of transaminase reactions that were not ATP-dependent. This problem has previously been reported in an atom mapping benchmark study prior to the publication of RXNMapper (Preciat Gonzalez et al, 2017). These transaminase reactions were identified in all metabolic networks and manually atom mapped based on the KEGG reaction mappings (Fig EV6).

Pre-processing genome-scale metabolic models

Several pre-processing steps were required to compute AEFMs from the GEMs. The stoichiometric coefficients of external reactions were set to one and their fluxes rescaled to carry equivalent units of source and sink metabolites. All reactions containing pseudometabolites with no known SMILES strings were removed from the stoichiometry matrix and replaced with unimolecular source and sink reactions carrying equivalent units of flux. The same procedure was also applied to reactions containing non-integer stoichiometries. These pseudometabolites were next removed from the stoichiometry matrix rows, and unimolecular reactions involving the same external or internal metabolites were aggregated. Finally, we had to remove a small number of reactions whose reaction SMILES strings exceeded the 512 token limit of RXNMapper. These reactions were likewise converted into unimolecular source and sink reactions to maintain steady state flux across the network. We note that all pre-processing steps described above are wrapped into convenience functions within

our Julia package MarkovWeightedEFMs.jl.

Benchmarking enumeration of atomic-versus-standard EFM

We benchmarked our Julia package MarkovWeightedEFMs.jl (version X.0.0) running Julia version 1.10.2 to enumerate atomic EFMs versus the molecular EFM enumeration program FluxModeCalculator (van Klinken & Willems van Dijk (2016)) running MATLAB version R2018a with default parameters including network compression. We chose FluxModeCalculator because it is the state-of-the-art program for enumerating molecular EFMs. Benchmarks were performed on a 64-bit Linux operating system with 64GB of memory and an AMD 5950X CPU with 16 cores/32 threads. All benchmarks reported the wall time to run the atomic or molecular EFM enumeration functions and did not include startup times to open Julia/MATLAB, precompile Julia functions, or load packages and input data. A benchmark time of 'did not finish' (DNF) was assigned if either program could not complete EFM enumera[illegible] practice, only FluxModeCalculator failed to enumerate the molecular E[illegible]atasets.

Ask self if this is all needed or could be shortened?

Model preliminaries

The structure of a knowledge-driven metabolic flux network is encoded as an m by n stoichiometry matrix S with m distinct metabolites and r irreversible reactions. The fluxes along each biochemical reaction v are positive, real-valued weights that denote the number of times each reaction is occurring per unit time. Under metabolic steady state, the net fluxes consuming and producing each metabolite are balanced and $S \cdot v = 0$. These reactions may be unimolecular–single substrate to single product transformations– or involve multiple substrates, products, or stoichiometric copies of participating metabolites. We do not require the network to be strong connected; however, our method will treat independent components separately. In contrast with our previous unimolecular

CHMC model, we impose additional requirements on the input metabolites and reactions. The 3D structure of each metabolite must be known and encoded as a SMILES structure. We recommend, where applicable, using isomeric SMILES structures to avoid ambiguous or incorrect atom mappings. We further require integer-valued stoichiometric coefficients because we consider atoms to be indivisible entities from which we enumerate their AEFMs. We lastly require the input metabolic flux network to be open such that external fluxes enter and leave via source and sink metabolites. We note this is generally not a problem in large-scale metabolic networks; our requirement is not a technical limitation of our method but enforced to promote the enumeration of all source-to-sink AEFMs remodelling nutrient sources.

A knowledge-driven metabolic network can be decomposed into a finite set of K molecular EFMs. In a unimolecular reaction network, the metabolites and steady state fluxes can be represented as a graph and the molecular EFMs correspond to weighted simple cycles carrying steady state flux through the metabolite vertices. In a network containing multispecies reactions, the correspondence between molecular EFMs and simple cycles is lost as molecular EFMs may include higher order reactions involving flows across multiple substrates and products. For either type of knowledge-driven metabolic network, their observed steady state fluxes can always be represented as a positive, linear combination of their K molecular EFMs. This flux decomposition problem can be represented by the system of linear equations $Aw = f$, where A is an r by K matrix of zero or natural numbers denoting the number of times each reaction occurs molecular EFM $k = 1, 2, \ldots, K$, where w_k is the molecular EFM weight. When the number of molecular EFMs exceeds the number of linearly independent fluxes, w is underdetermined and there are an infinite number of ways to assign molecular EFM weights to explain the observed, steady state fluxes.

Markovian decomposition of fluxes onto EFM weig

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In our previous work (Chitpin & Perkins, 2023), we proposed Markovian EF[illegible] as a means to resolve the flux decomposition problem in networks consisting exclusively of unimolecular reactions. We formulated a probabilistic model of single particle flux, in the form of a Markov chain, where we assumed the probability of this hypothetical particle transitioning from one metabolite state to another was proportional to the observed, steady state flux. We further proposed a way of efficiently computing those molecular EFM probabilities and weights by introducing the *cycle-history Markov chain* which both enumerated the molecular EFMs and uniquely identified their probabilities which were rescaled to weights based on the absolute flux magnitudes. To generalize this CHMC method to networks containing multispecies reactions, we introduce the concept of AEFMs which can be enumerated and their weights computed from an atomic variant of our CHMC method (referred to as an atomic cycle-histor[illegible] intuitive justification of our metl[illegible]ithm correctness in our original p[illegible]

Following Chitpin & Perkins (2024), we developed... => Try to make all this into one-paragraph high-level summary, and point to code!

Single atom, discrete-time Markov chain

We developed a probabilistic model of atomic flux to compute atomic EFM weights in steady state metabolic flux networks. At this atomic level, individual atoms within metabolite undergo biochemical reactions and move from the substrate to product side of each reaction equation. As the molecules within metabolic flux network are stoichiometrically balanced under metabolic steady state, so too are the corresponding atomic fluxes.

For a user-specified atom within a given source metabolite, we identify all possible metabolite/atom states that are traversable by that initial state and constrained by the reaction atom

mapping predictions. For reactions involving single stoichiometric copies of each substrate and product, the atom of interest will always move from a single substrate to product molecule. If this atom is present in a substrate with multiple stoichiometric copies, we assume there is an equal probability of that atom occurring in either copy. Depending on the substrate stoichiometric copy, that atom may move to (i) different product metabolite/atom states, (ii) the same product but at different atomic positions, or (iii) the same metabolite/atom position. In all cases, we set the atomic flux of the atom transitioning from either stoichiometric copy equal to the reaction fluxes to maintain atomic mass conservation (Expanded Views 5).

For a given source metabolite/atom state, the subsequent atomic transitions are encoded as an atomic transition graph $G = (V, E, f)$, where V is the set of nodes corresponding to metabolite/atom states, E is the set of directed edges corresponding to reactions that move the atom from one metabolite to another, and $f \equiv v$ are the atomic mass fluxes. We assume there exists a special node $v_{ext} \in V$ which represents the external environment. Source flux entering the graph occurs along the edge $e = (v_{ext}, r)$, where $r \in V - v_{ext}$ is the source metabolite/atom state. Sink fluxes leaving the graph occur along edges $e = (i, v_{ext})$, where $i \in V - v_{ext}$ are the sink metabolite/atom states. The introduction of v_{ext} closes off the graph and ensures it is strongly connected, where there exists a path in G from any node i to any other node j.

We make a Markovian assumption that the movement of an atom from metabolite/atom position to another in G is proportional to the observed steady state (atomic) flux of that reaction. We denote these transition probabilities $T_{ij} = f_{ij} / \sum_{j'} f_{ij'}$ for $i \neq j \in |V|$. From here, we follow the same intuition as our unimolecular CHMC algorithm (Chitpin & Perkins, 2023) to model atomic flows within the network. Starting from an initial source metabolite/atom state s_0, the atom of interest will transit a random sequences of states $s_0, s_1, s_2, \ldots$. As G is strongly connected, that sequence will, with probability of one, include all states infinitely many times. If the atom visits state i, it

i is state
not time index.
Use state i at time t
(so s_t = i)

will eventually return to state i. The sequence of states in [illegible] itutes a cycle, albeit not necessarily a simple cycle. The sequence many inclu[illegible]ested or interleaved simple cycles. However, this cycle can always be decomposed into a unique set of simple cycles by repeatedly removing the earliest simple cycle found within the sequence, which we refer to as the "first closure reduction rule". The AEFMs are the set of all simple cycles in G. Thus, we compute the AEFM weights by setting them proportional to the steady state probabilities of their corresponding simple cycles.

Atomic cycle-history Markov chain

We compute the steady state probabilities of each AEFM following our previously described CHMC method for networks of unimolecular reactions (Chitpin & Perkins, 2023). Briefly, the ACHMC is a special type of Markov chain, where we expand the notion of state to include sequences of metabolites transited by the atom. The ACHMC therefore traces the history of a given atom on its way to completing a simple cycle corresponding to an AEFM.

We first initialize the ACHMC on source metabolite atom s_0 within G. Although we could technically choose any arbitrary metabolite/atom state for s_0, it is much easier to interpret AEFMs rooted on a source metabolite flowing into the network. Every state S in the ACHMC corresponds to a simple path $s_0, s_1, s_2, \ldots$. If a reaction transforms s_m to s_{m+1}, the corresponding state $S \equiv s_0, s_1, \ldots, s_m$ transitions to the longer path $S' \equiv s_0, s_1, \ldots, s_m, s_{m+1}$. If a reaction transforms $s_{m+1} = s_i$, the corresponding state $S \equiv s_0, s_1, \ldots, s_m$ transitions to the shorter path $S' \equiv s_0, s_1, \ldots, s_i$ For either case, the ACHMC transition probability is $\mathcal{T}_{S,S'} = T_{s_m, s_{m+1}}$.

Computing steady state AEFM probabilities

Once constructed, the ACHMC describes all possible simple cycles that a given source metabolite atom can transit through. These simple cycle probabilities are computed by analyzing the steady state dynamics of the ACHMC. As the atomic transition graph G is strongly connected, the corresponding CHMC will also be strongly connected. There thus exists a stationary distribution π which corresponds to the steady state probability of the atom occupying each CHMC state.

To compute AEFM probabilities, we first identify all transitions E_K in the ACHMC that close simple cycles corresponding to EFM k. For source-to-sink AEFMs, there exists only a single simple cycle and therefore only one corresponding simple cycle. For internally looped AEFMs, there may exist multiple simple cycles starting at different metabolite/atom states, counting towards the same AEFM. We define the steady state probability of transiting EFM k as the sum of the steady state probabilities of those transitions or:

$P_k = \sum_{(S,S')\in E_k} \pi_S \cdot \mathcal{T}_{S,S'}$.

As the AEFM probabilities are proportional to the atomic fluxes, these probabilities may be scaled to weights based on the absolute magnitude of the fluxes. Because we enforce that each source metabolite is produced by an input reaction with stoichiometry of one, the sum of all corresponding source-to-sink AEFMs must be proportional to the input metabolite flux. If $l \in K$ denotes all source-to-sink AEFMs flowing through a given source metabolite m_0 and f_0 is the input flux producing that metabolite, we define [illegible] constant $\alpha = f_0/\sum_l P_l$. This scaling not only ensures the total input/output [illegible] tructed, but also that every individual flux in the network is correctly expla[illegible] of AEFM weights.

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Details associated with computing the (A)CHMC stationary distribution

Following basic Markov chain theory, the stationary distribution of the CHMC, π is proportional

to the eigenvector of the CHMC transition probability matrix. The eigenvector can be solved by direct or iterative numerical methods. The choice of algorithm becomes increasingly important as the size of the CHMC transition matrix grows exponentially in highly-connected, large-scale networks. We note that iterative eigenvector solving algorithms from some packages (e.g. Arnoldi method from Arnoldi.jl, GMRES from IterativeSolvers.jl) resulted in negative eigenvector coefficients or NaN values. These are possibly numerical errors and occurred more frequently in larger CHMC matrices. Repeatedly re-running these methods sometimes resulted in a correct eigenvector, albeit with significantly longer computation times. We observed that the linear solver from the Julia package LinearSolve.jl efficiently solved for the correct eigenvector without error using the default parameters. Hence, we use and recommend this package to compute the stationary distribution of large ACHMCs.

Data availability

The code and data to reproduce this study are available at

https://www.github.com/jchitpin/reproduce-efm-paper-2024.

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Author contributions

Justin G. Chitpin: Conceptualization; data curation; methodology; software; formal analysis; investigation; visualization; writing - original draft; writing - review and editing.

Theodore J. Perkins: Conceptualization; methodology; supervision; funding acquisition; investigation; project administration; writing - review and editing.

Disclosure and competing interests statement

The authors declare no competing interests.

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Figure legends

Fig 1: Atomic cycle-history Markov chain (ACHMC) pipeline. (a) An example molecular EFM in glycolysis. (b) An example AEFM tracing the flow of a given glucose carbon in glycolysis. (c) The major steps within our pipeline to enumerate AEFMs and compute their weights.

Fig 2: AEFMs are computationally tractable compared to molecular EFMs in five GEMs. (a) Number of atomic and molecular EFMs in the five GEMs. (b) The run time of our MarkovWeightedEFMs.jl method versus FluxModeCalculator running in serial. (c) Serial run time scaling of

MarkovWeightedEFMs.jl as a function of the number of atomic EFMs. (D) Speedup of selected ACHMCs as a functional of additional threads.

Fig 3: Structural analysis of AEFMs in five genome-scale metabolic models. (a-b) Categories of carbon and nitrogen EFMs. (c-d) Distribution carbon and nitrogen AEFM lengths. (e-f) Ratio of the greatest and smallest number of AEFMs between carbons and nitrogens within the same source metabolite. Representative metabolites from each dataset are labelled; labels with asterisks indicate that metabolite is also present in other networks.

Fig 4: Carbon AEFM weight analysis of the HepG2 metabolic flux network. (a) A small number of highly active AEFMs explain the majority of glutamine flow. (b) Subnetwork of the top five AEFMs across each glutamine carbon collapsed onto metabolites. (c-e) Sankey diagrams of the top five glutamine carbon AEFMs. (f) Schematic mapping AEFMs onto traditionally defined metabolic pathways/subsystems. All metabolite names are labelled following the BiGG database nomenclature.

Tables and their legends

Table 1: Summary statistics for the five metabolic networks in this study.

Table 2: Number of metabolites and reactions removed from each network during pre-processing.

Expanded View Figure legends

Expanded View 1: Network visualization of the five GEMs after pre-processing. Metabolites (non-grey nodes) are connected to reaction entities (grey nodes) with node sizes proportional to their degree. Graphs were constructed in Gephi version 0.10 using the Yifan-Hu proportional network layout with default parameters. (a) E. coli core. (b) iAB RBC 283. (c) iIT341. (d) iSB619. (e)

HepG2.

Expanded View 2: FluxModeCalculator scaling across multiple threads.

Expanded View 3: Most ACHMCs traverse less than 10% of the total metabolite-carbon/nitrogen state space. Each point corresponds to an ACHMC rooted on a given source metabolite carbon or nitrogen.

Expanded View 4: Cumulative explained carbon mass flow of input glucose and amino acid carbons from the HepG2 model. The cumulative explained mass flow is truncated at 99% to avoid displaying AEFMs that carry little atomic mass flow.

Expanded View 5: Rules for assigning atomic fluxes from atom-mapped reactions with biological examples. Schematic node colours represent ground truth atom mappings across substrate(s) and product(s). Node labels correspond to distinct atomic indices of substrate and product stoichiometric copies. (a) For reactions with single stoichiometric copies of substrates and products, each atom within a given substrate moves to a single atomic position within a product with atomic flux equal to the reaction flux. (b-d) For reactions with s stoichiometric copies of a given substrate, there are s copies of each metabolite-atom state on the substrate and product side of the reaction. The probability of any atom within a substrate stoichiometric copy moving to a product metabolite-atom state remains equal to the reaction flux. (b) The stoichiometric copies of the substrate fuse together into a single product. (c) The stoichiometric copies of the substrate form distinct products. (d) The s stoichiometric copies of the substrate form s copies of products with identical atomic backbones. (e) Biological example of rule a where the atoms in each substrate map to a single position on the corresponding product. (f) Biological example of rule b where two stoichiometric copies of glutathione fuse together to form a single stoichiometric copy of glutathione disulfide. (g) Biological example of rule c where two stoichiometric copies of acetyl-CoA undergo a reaction to produce a stoichiometric copy of two distinct products.

Expanded View 6: Examples of incorrectly atom-mapped transaminase reactions present in one or more of the five networks. (a) Phenylalanine transaminase. (b) Isoleucine transaminase. (c) Tyrosine transaminase. (d) Aspartate transaminase.